

HireFlow

An AI-native, multi-tenant HRMS for medium and large enterprises · Product Proposal

Product Vision

HireFlow moves enterprise HR from reactive administration to proactive, intelligence-driven decision-making. Our thesis: every HR workflow that produces structured output — a scored resume, a policy answer, a performance summary, a sentiment signal — should have a specialised AI agent embedded as an opinionated co-processor, not as a bolt-on chatbot. The platform ships with **eight production AI agents** across five HR modules, all observable through a real-time cost, latency, and accuracy console — making AI adoption measurable from day one, which is the prerequisite for enterprise procurement.

Client Problem

Enterprise HR teams face three compounding problems. **Volume without signal:** hundreds of applications per role, thousands of policy questions per month, dozens of review cycles per year — capacity to process this carefully does not scale linearly with hiring. **Compliance blind spots:** biased JD language, inconsistent policy answers across HR reps, undocumented review rationale — every one of these creates legal exposure that grows with headcount. **AI adoption opacity:** enterprises that pilot AI for HR cannot answer 'what does this cost per call, is the output accurate, which team drives usage?' — without that visibility, AI stays in pilot forever.

Target Users / Personas

Persona	Pain Point	HireFlow Value
CHRO / HR Director	Budget owner; no visibility into AI ROI	AgentOps console: cost/call, latency, agent breakdown
HR Admin	Manual screening; inconsistent policy answers	AI scoring · PolicyBot RAG · review synthesis
Hiring Manager	Slow pipeline; no signal on who to interview	Kanban + AI scores + targeted interview questions
Employee	Waits hours/days for basic policy answers	PolicyBot: instant, sourced, 24/7 answers
Platform Admin (SUPER_ADMIN)	Multi-tenant provisioning + cross-company audit	Full cross-company visibility + AgentOps

Business Assumptions

- **Buyer:** HR Director or CHRO at a 200–5,000-person company, replacing legacy HRMS or augmenting one with an AI layer.
- **Pricing:** SaaS, per-seat, tiered by headcount; AI token cost passed through with margin; AgentOps data justifies the premium.
- **Sales cycle:** 3–6 months; PoC on a demo tenant before contract.
- **Compliance baseline:** GDPR-aware data handling assumed; SOC 2 Type II is the next enterprise gate.
- **AI cost viability:** PolicyBot < \$0.01/question at scale via prompt caching (\$3/M in, \$15/M out, ~90% cache-hit rate on the same policy corpus).

AI-Native Capabilities (8 production agents)

Agent	Trigger	Structured Output / Pattern
JD Parser	Job post create	Required skills, experience years (tool use)
Bias Detector	JD save / on-demand	Flagged phrases, DEI alternatives, risk LOW/MED/HIGH
Resume Screener	Candidate upload	Score 0–100, strengths, skill gaps
Interview Generator	Candidate detail	3–5 targeted questions from identified gaps
PolicyBot	Employee question	Sourced answer + confidence + auto-escalate (RAG + caching)
Review Synthesizer	HR triggers synthesis	Strengths, growth areas, narrative — one summary per employee
Survey Insight	Survey results view	Keyword themes from anonymous responses
Analytics Narrator	Analytics page load	Natural-language summary of workforce metrics

Discipline: Anthropic *tool use* is used everywhere typed data is needed — no regex parsing of free text. **Prompt caching** applied to stable system prompts and policy contexts. **Full observability:** every Claude call passes through `observedAiCall()`, which captures input/output tokens, latency (ms), and cost (USD) and writes them fire-and-forget to `AiCallLog` for the SUPER_ADMIN AgentOps console.

Scalability & Enterprise Readiness

- **Multi-tenant by design:** `companyId` on every model + every Prisma query — cross-tenant access is architecturally impossible.
- **Stateless serverless API:** Vercel functions scale horizontally and to zero; Supabase transaction pooler handles cold-start fan-out.
- **Soft deletes only:** status fields (ACTIVE/INACTIVE, OPEN/CLOSED) preserve full audit history for compliance forensics.
- **AuditLog** table records every user action with entity, action type, and diff — enables forensic review and regulatory export.
- **AI cost control:** AgentOps surfaces per-company AI spend, enabling SaaS chargeback or internal cost allocation.

Security & Compliance Assumptions

AuthN: NextAuth 5 JWT sessions with `HttpOnly` cookies and short TTL (production would delegate to an IdP via SAML/OIDC). **AuthZ:** four-layer RBAC — middleware (route), API handler (role → 403), data queries (`companyId` filter), and UI (RoleGate + sidebar filtering). **Data isolation:** every table carries `companyId`; no query can accidentally cross tenants without explicit code change. **Secrets:** all keys in encrypted Vercel env vars — zero secrets in repo. **Transport:** TLS end-to-end (Vercel HTTPS + Supabase SSL). **AI data:** resume / policy text is sent to Anthropic — enterprises requiring data residency would use Anthropic Enterprise tier or swap `lib/ai/client.ts` to an on-prem model. **Roadmap:** SOC 2 Type II, GDPR right-to-erasure flows, SSO/SAML via Okta or Azure AD.

Modules Intentionally Not Built

Module	Reason Deferred
Payroll	Bank-grade security + country-specific tax rules + payment-processor integrations exceed prototype scope.
Benefits Admin	Carrier integrations and jurisdiction-specific legal requirements; high complexity, low prototype signal.
Time & Attendance	Requires hardware (badges, biometrics) — no meaningful prototype without physical layer.
SSO / SAML	Table stakes for production but adds no AI-differentiation value in a prototype.
Email / Webhooks	Deferred in favour of in-app UI; avoids SMTP / vendor schema dependencies.
Native mobile app	Web prototype + responsive layout; native UX deferred.

Key Trade-offs Made

Decision	Trade-off / Rationale
Monolith over microservices	Single deploy, no network overhead, fast iteration — right call for prototype.
Keyword RAG (no vector DB)	Lower retrieval accuracy at >100 docs, but works well for typical policy corpus; pgvector swap is an isolated change.
Fire-and-forget AI logging	Log writes can fail silently — acceptable because observability is non-critical-path; user latency is preserved.
JSON-stringified arrays in Prisma	Less directly queryable in SQL, but schema simplicity wins at prototype scale.
prisma db push over migrations	No migration history — fine for rapid iteration; production switches to <code>prisma migrate</code> .